Arunima Anand

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EDUCATION

M.S. in Data Science | University of California, San Diego — Sep 2025 – Mar 2027 | San Diego, CA
3.97 GPA | **Available to Start: April 2027**
Relevant Coursework: ML Systems, Recommender Systems, Scalable Data Systems, Causal Inference, Statistical Models, Responsible ML

Bachelor's in Economics (Minor in Mathematics) | University of Delhi — Jul 2019 – Jul 2022 | Delhi, India
3.88 GPA (First Division with Honors)

PROFESSIONAL EXPERIENCE

Applied Science Intern (Measurement & Identity) | The Trade Desk — Jun 2026 – Sep 2026 | New York, NY

- Developed a prototype **probabilistic audience-reach estimator** over **16B+ ad events** using censored frequency modeling, **identity-graph churn signals, HyperLogLog sketches,** and **vectorized feature embeddings** for cohorts with limited identity coverage.
- Built **distributed SQL** and **PySpark pipelines** across ad group, campaign, and advertiser aggregation levels, validating reach estimates against independent signals using **temporal** and **entity holdouts**, sensitivity analysis, and **A/B tests**.
- Applied **knowledge distillation** to transfer signal from richer representations into **LightGBM models** using production-efficient features, enabling **sub-50 ms inference at scale**.
- Productionized estimator outputs through **Databricks APIs** integrated with the **trading platform**, powering an **advertiser-facing dashboard** for reach analysis, cohort diagnostics, and continuous model monitoring.

Data Scientist II (AI/ML Engineering) | Deloitte — Oct 2022 – Aug 2025 | Delhi, India

- Developed an **automated demand forecasting platform** across **7,000+ product-level time series**, engineering lag and rolling-window features and **benchmarking Prophet, XGBoost, LSTM,** and **ARIMA** with best-fit model selection, achieving **93.63% forecast accuracy** and **$5M+ in savings**.
- Productionized **scheduled training** and **batch inference workflows** on AWS SageMaker, S3, and Step Functions with automated feature generation, retraining, validation, and drift monitoring, reducing **end-to-end runtime** by **91%**.
- Designed and deployed an **ML framework** for **product scoring** and **classification** using feature-weighted aggregation, **distribution-aware binning,** Random Forest, and LightGBM, achieving **95% holdout accuracy** and maintaining production stability through **drift-aware recalibration** and **model versioning**.
- Architected an **NLP pipeline** to standardize inconsistent defect text and discover recurring patterns using **BERT embeddings**, **HDBSCAN clustering**, and association mining, mapping **31%** of records to **canonical labels** and improving detection yield by **1.6×**.

PROJECTS & RESEARCH

Personalized Ad Ranking for a Housing Marketplace — Oct 2025 – Dec 2025

- Built a **personalized ad** and **listing ranking system** over **500K+ clickstream events** and **100K+ user profiles**, leveraging behavioral features and user-item affinity signals to rank property listings and campaign content.
- Developed **user-level affinity scores** for personalized ad targeting and **high-affinity audience identification**, improving campaign planning signal quality by **32%**.

Evidence-Grounded LLM Evaluation & Retrieval — Jan 2026 – Mar 2026

- Designed a distilled **LLM-as-a-judge** with **source-evidence retrieval**, using **PyTorch** and **LoRA** to adapt **Qwen2.5-3B** for summary evaluation across coherence, consistency, fluency, and relevance.
- Conducted fine-grained meta-evaluation against **human annotations**, achieving **75.5% off-by-1 agreement** and identifying a **ranking-calibration tradeoff** in automated evaluation.

Computer Vision + LLM Food Waste Intelligence Platform — Apr 2026

- Built a **CV** and **LLM platform** using **agentic AI coding workflows**, integrating S3, Lambda, AWS Rekognition, Grok Vision, and Databricks for **food-waste ingestion, deduplication,** and **classification**.
- Delivered **Prophet-based weekly waste forecasts** and analytics through **FastAPI** and **React** to support restaurant inventory planning.

SKILLS

- **Programming**: Python, SQL, Java, C++, R
- **Libraries & Frameworks:** PyTorch, TensorFlow, PySpark, scikit-learn, NumPy, Pandas, statsmodels, Matplotlib, NLTK
- **Data Science & Machine Learning:** Probabilistic modeling, statistical & causal inference, A/B testing, recommender systems, ranking, identity-graph analysis, classification, regression, feature engineering, time-series forecasting, XGBoost, LightGBM
- **Deep Learning, Retrieval & GenAI:** Knowledge distillation, embeddings, LSTMs, transformers, BERT, NLP, information retrieval, TF-IDF, semantic search, LLM evaluation, RAG, prompt engineering
- **MLOps, Cloud & AI Engineering:** Distributed computing, Databricks, AWS (S3, Lambda, Step Functions, SageMaker), FastAPI, GitHub, CI/CD, real-time serving, model monitoring and versioning, drift detection, AI-assisted development